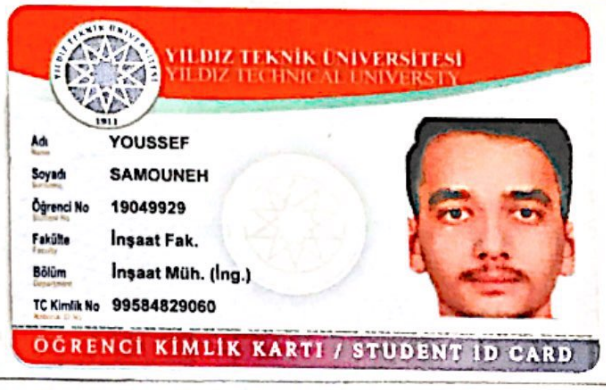
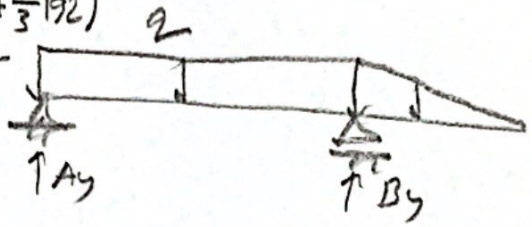
Name: YOUSSEF
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ID No: 19049929
Group: 5



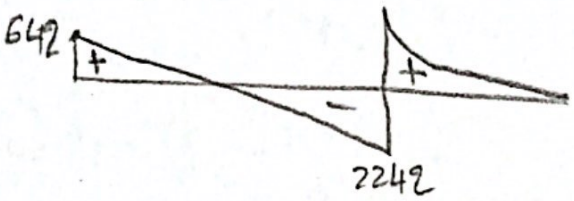
$L = 192\,cm \qquad \sigma = 140\,MPa \qquad \tau = 80\,MPa$

$\Sigma M_A = 0;\ B_y = \dfrac{2(192)\cdot\frac{192}{2} + \frac{2(192)}{2}(192+\frac{1}{3}192)}{192}$



$B_y = 96q + 128q = \boxed{224q}$

$A_y = 2(192) + \dfrac{2(192)}{2} - 224q = 64q$



$x_0 = \dfrac{64q}{2} = 64\,m$

$max\,M = \dfrac{(64q)^2}{2\cdot q} = \boxed{2048q}$

$I = \dfrac{8\cdot 14^3}{12} - \dfrac{6\cdot 10^3}{12} = 1329{,}3\,cm \qquad S_x = 2\cdot 8(6) + 2(1)(5)(\frac{5}{2}) = 121\,cm^3$

$\sigma = \dfrac{2048q}{1329{,}3}(7) \le 14000 \qquad \boxed{q = 1298\,N}$

$\tau = \dfrac{224q\cdot 121}{1329{,}3\cdot 2\cdot 1} \le 800 \qquad \boxed{q = 784{,}7\,N}$

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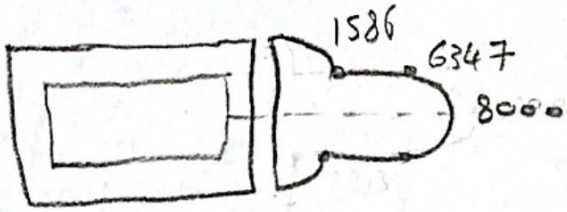
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Group: 5



$S_1 = 2 \cdot 8 \cdot 6 = 96 cm^3 \quad S_2 = 121 cm^4$

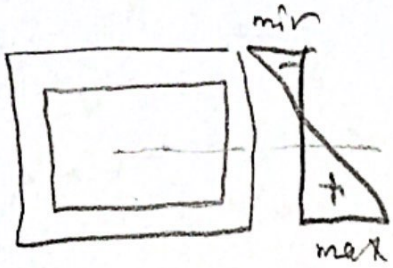
$$\tau = \frac{(224)x}{1329,3 \times 8} = \frac{784,7 \times 96}{1329,3 \times 8} \simeq 1586 N/cm^2 \qquad \tau = \frac{(224)x}{1329,3 \times 2 \cdot 1} = \frac{784,7 \times 96}{1329,3 \times 2 \cdot 1} \simeq 6347 N/cm^2$$

$\tau = 8000$



$$\sigma_{max} = \frac{2048(784,7)}{1329,3} \times 7 = 8462,6$$

$$\sigma_{min} = \frac{2048(784,7)}{1329,3}(-7) = -8462,6 N/cm^2$$



$$\sigma_E = \frac{2048(784,7}{1329,3}(-5) = -6044,78 N/cm^2$$

$\tau_E = 6347 N/cm^2$

$$\sigma_{1,2} = -\frac{6044,78}{2} \pm \sqrt{\left(\frac{6044,78}{2}\right)^2 + 6347^2}$$

$\sigma_1 = 4007,5$

$\sigma_2 = -10052,8$

$|4007,7 - (-10052,3)| \leq 14000 \Rightarrow 14060 > 14000$

unsafe

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